

The Energy Balance Project

Homeostatic Field Model and Local Actor-Motion Law

Foundation Document — Version 2.3 (Audit of Foresight and Regeneration)

Status: two controlled stages completed. Stage 1 (V2.2.1 audit) isolates the cause of the V2.2 clustered-world failure. Stage 2 (V2.3) adds regeneration and tests whether immediate homeostatic improvement destroys future regenerative capacity. All claims are backed by scripts and by a 19-test suite.

Two headline results, both counter to the naive expectation. (1) The clustered failure is driven mainly by **myopia, not dissipation**: lossless transport ($\eta=1$) lifts clustered viability only 51→57%, while adding foresight ($H=10$) lifts it 54→80%. (2) In a closed regenerative (Allee) economy, a longer foresight horizon of the specified single-action form does **not** protect regeneration — it **destroys** it. Myopic rules keep every source alive through a shock; naive $H=10$ drives all 32 sources below their Allee threshold to irreversible collapse. The cause is a systematic bias in the single-action counterfactual.

Abstract

V2.2 left one open question (why clustered sources fail) and one overclaim (“strict dominance”), both corrected here. We first audit the clustered failure by separating three possible causes — layout variance, transport dissipation, and short-horizon myopia — and find myopia dominant. We then add the four V2.0 source behaviours (external flow, logistic stock, Allee stock, finite stock) and pose the project's central question in a closed economy whose only supply is regeneration. Contrary to expectation, the regeneration-aware actor of V2.0 Section 9 (a longer local counterfactual) is biased toward over-harvest: because it evaluates a single action in isolation, longer horizons credit a harvest more (the source appears to regenerate freely and the fed cell appears permanently rescued), so it over-harvests and drives Allee sources below their recovery threshold after a disturbance. Short-horizon rules are accidentally protective. This empirically validates the V2.0 design note (Sec. 7.2) that a critical regeneration threshold belongs in the burden and regeneration dynamics, not in a longer isolated lookahead.

1. Corrections carried from V2.2

Before new work, the V2.2 claims were corrected: “strictly dominates” became “lower burden and unmet demand overall, but not a strict Pareto improvement” (clustered viability is tied, 51.0% vs 51.6%, and the safe rule uses slightly more transport on the checkerboard); “scale-invariant” became “approximately size-insensitive across the three sizes tested”; the falsification table was reflowed to remove overlapping text; and the clustered failure was left attributed to “locality, delay, and dissipation” pending this audit.

2. Stage 1 — Audit of the Clustered Failure (V2.2.1)

Using the same engine and the same three worlds, only the decision rule changes. Three questions, each isolating one factor.

2.1 Is the failure robust across layouts?

Over 30 seeds on the random world, the safe rule beats the gradient rule robustly ($85.6\% \pm 10.1$ vs $71.7\% \pm 8.2$ viable). The clustered numbers below have zero variance because the current builder places the producer block deterministically; the 30 clustered “seeds” are therefore identical. The clustered conclusion rests on one layout plus the sweeps in 2.2–2.3, not on 30 distinct layouts — a limitation fixed in future work by randomising the cluster.

World	Model	viable % (mean $\pm$ sd over 30 seeds)
random	gradient	71.7 ± 8.2
random	safe	85.6 ± 10.1
clustered	gradient	51.6 ± 0.0 (deterministic layout)
clustered	safe	51.1 ± 0.0 (deterministic layout)

2.2 Is it transport dissipation? (vary eta)

On the clustered world, making transport more efficient barely helps. Even perfectly lossless transport recovers only a few points of viability:

transport efficiency eta	0.90	0.95	0.98	1.00
clustered viable %	47.1	51.1	54.7	57.3

2.3 Is it myopia? (vary horizon H)

Adding foresight helps a great deal. Moving from a one-tick to a ten-tick local counterfactual lifts clustered viability from 54% to 80% (a wider evaluation radius adds nothing beyond H, so horizon length, not radius, is the active variable):

foresight horizon H	1	3	10	30	10 (radius 4)
clustered viable %	54.0	62.2	80.1	78.2	80.1

V2.2.1 audit: isolating the cause of clustered failure

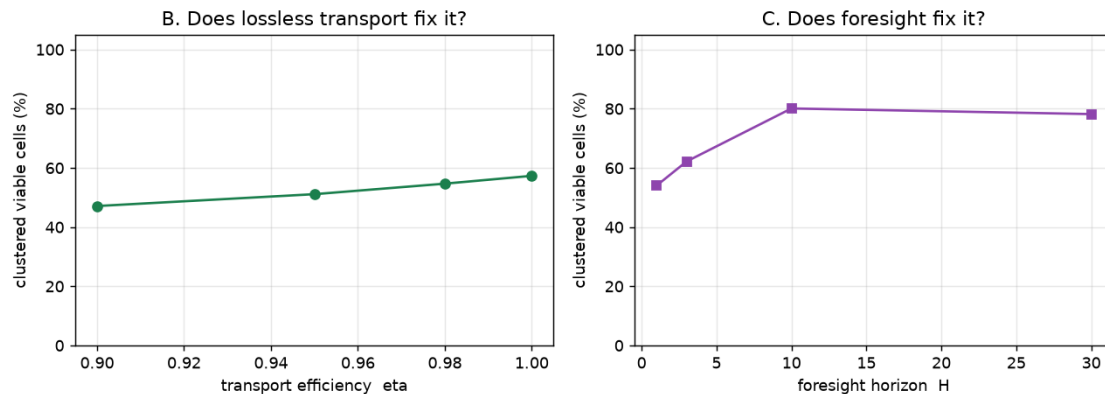


Figure 1. Isolating the clustered failure. Lossless transport (left) barely helps; foresight (right) helps substantially. The failure is chiefly myopia: the one-tick safe rule will not move capacity through neutral intermediate cells toward a distant deficit.

Stage 1 conclusion. The V2.2 clustered failure was principally short-horizon myopia, not transport loss. This corrects the V2.2 wording, which had left dissipation as a candidate cause.

3. Stage 2 — Regeneration (V2.3)

3.1 Four source behaviours

The four V2.0 source classes are expressed through existing parameters (s , ρ) plus an Allee threshold A_i , with the natural-update ledger extended to accept signed regeneration:

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external flow :  $s_i > 0$ ,  $\rho_i = 0$ 
logistic stock:  $g_i(x) = \rho_i \times (1 - x/K_i)$  ( $A_i = 0$ )
Allee stock    :  $g_i(x) = \rho_i \times (1 - x/K_i) (x/A_i - 1)$  ( $A_i > 0$ )
finite stock   :  $g_i(x) = 0$ 
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Allee note: $g_i(x) < 0$ for $0 < x < A_i$ (a source declining below its threshold)

3.2 The Allee trap and the central question

A closed economy (no external inflow) on an 8x8 checkerboard: producer cells are Allee stocks, consumer cells have demand. The only supply is regeneration. Crucially $A = 8$ while the viable floor $L = 4$, so a source can sit inside the viable band ($x \geq L$) yet below its regeneration threshold ($x < A$) and be dying. The question:

Does immediate homeostatic improvement destroy future regenerative capacity?

We compare the raw gradient, the safe $H=1$ rule, and the regeneration-aware Model D at $H = 3, 10, 30$. A shock at tick 500 knocks every source down by 55%, landing them just above A , to test which rule lets them recover.

3.3 Result: longer naive foresight destroys regeneration

Model	viable % (end)	viable % (2nd half)	source stock	dead sources	cum. unmet	disc. burden
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gradient	100.0	100.0	585.8	0 / 32	0	65,881
safe (H=1)	100.0	100.0	587.1	0 / 32	0	65,352
horizon H=3	100.0	100.0	584.7	0 / 32	0	63,539
horizon H=10	0.0	19.2	0.0	32 / 32	6,658	224,105
horizon H=30	37.5	48.8	132.8	24 / 32	4,557	165,333

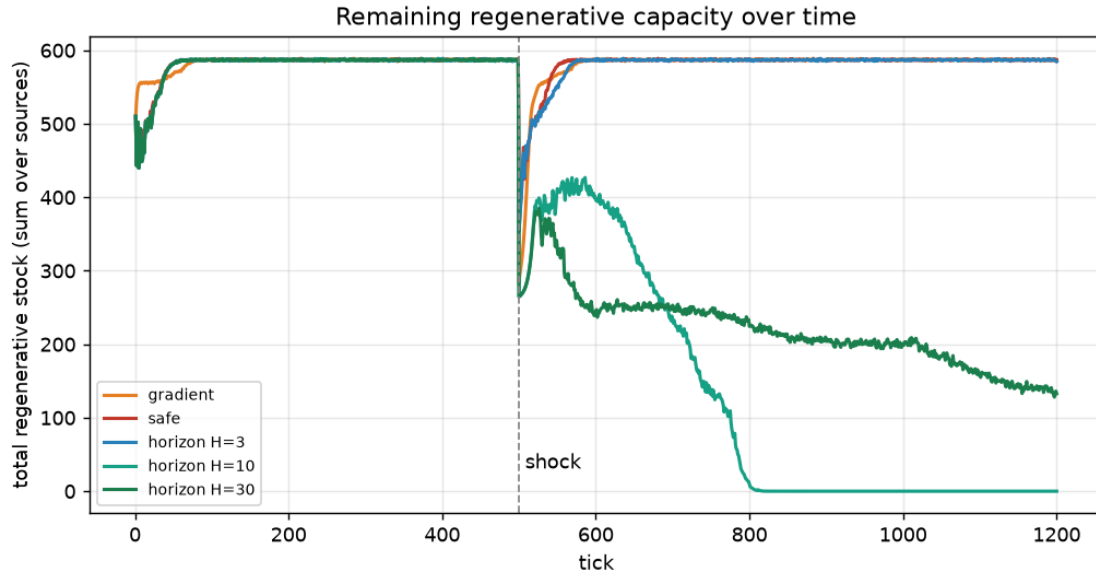


Figure 2. Remaining regenerative stock. All rules coexist while sources are healthy. After the shock, myopic rules (gradient/safe/H=3) let sources recover above the Allee threshold; naive H=10 keeps harvesting and drives them to zero (irreversible), H=30 bleeds out slowly.

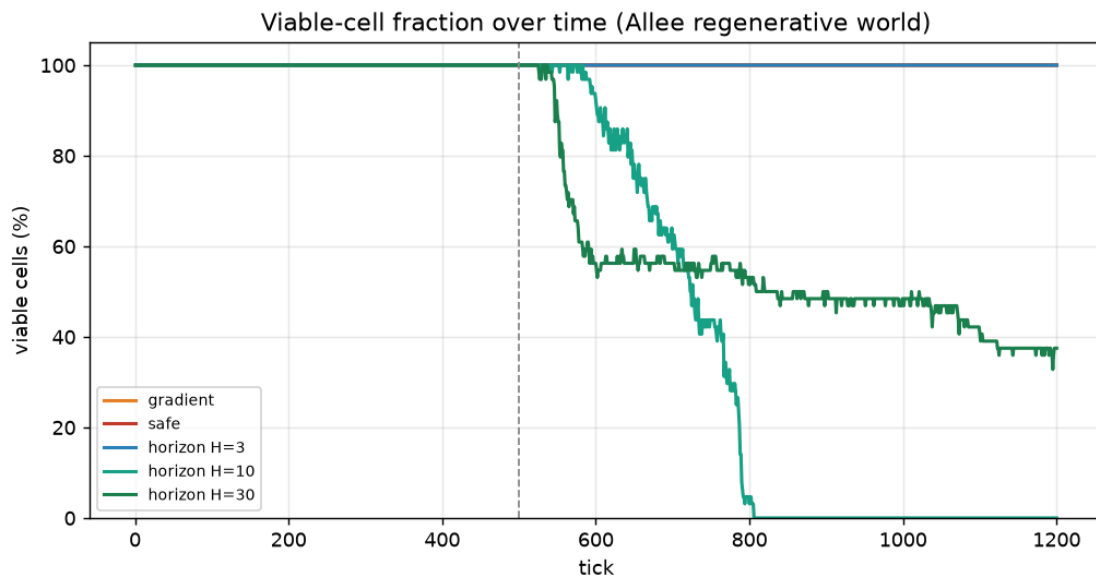


Figure 3. Viable-cell fraction. Short-horizon rules hold 100% through the shock; long-horizon rules collapse during recovery.

3.4 Mechanism: the single-action counterfactual over-credits harvest

The reversal is not a bug. The Section 9 counterfactual evaluates one action in isolation, assuming no other actions before or after. In a closed economy every actor harvests every tick, so this assumption is violated in two directions: the no-action branch over-states the fed cell's future deficit (it will in fact be re-supplied next tick), and the action branch lets the harvested source regenerate freely (it will in fact be re-harvested). Both biases grow with the horizon. A controlled single-harvest measurement shows the

predicted impact rising steeply with H, so longer horizons accept ever-more-aggressive harvesting:

horizon H	1	3	10	30
predicted impact of one harvest	0.00	+0.14	+34.2	+104.8

Near the Allee threshold this is fatal: after the shock, sources sit just above A; the myopic rules harvest gently (harvesting a low source carries high immediate burden, so they back off) and the sources climb back above A, whereas the long-horizon rule harvests hard and pushes them below A into irreversible decline. Short-horizon caution is here an accidental but effective form of regeneration protection.

4. Verdict

The engine is correct (ledger balances every tick, now including signed Allee regeneration; 19 tests pass across three suites). **The two stages answer the questions posed and both overturn the naive intuition.**

On foresight. Foresight is not uniformly good or bad. For pure transport/routing (clustered sources, no regeneration) a longer horizon helps: it lets capacity move through neutral cells toward distant deficits. For managing a regenerating resource, a longer horizon of the single-action form is harmful: it systematically over-credits harvest and destroys the resource after a disturbance.

On the central question. In this experiment immediate homeostatic improvement did NOT destroy future regenerative capacity — the naive long-horizon rule did. The correct reading is not “add foresight” but the V2.0 design note (Sec. 7.2): a critical regeneration threshold should be represented in the burden and regeneration dynamics (e.g. a penalty as x approaches A, or a sustainable-yield constraint), not delegated to a longer isolated counterfactual. The experiment is the evidence for that design choice.

Falsification criteria touched (V2.0 Sec. 13). “Short-horizon actors destroy long-horizon regeneration” was tested and, for the myopic rules, NOT triggered; the naive long-horizon rule triggered its mirror image. “Counterfactual costs more than it detects” is now concrete: the H-horizon evaluation is both expensive and, here, actively misleading.

5. Limitations and Next Steps

1. **Threshold-aware burden.** Add a penalty (or regeneration-reserve constraint) as x approaches A_i and re-run Stage 2; expected to give sustainable harvest without a long counterfactual. This is the priority.
2. **Fix the clustered layout.** Randomise cluster position/shape so the audit's variance claim rests on genuinely distinct layouts.
3. **Better counterfactual.** If a lookahead is kept, model repeated/aggregate actions (or a mean-field of other actors) so it stops assuming a single isolated harvest.
4. **Then EBU.** Only once a threshold-aware rule sustains regeneration should the EBU ledger be added, and tested for gaming.

Version status. V2.3 keeps the V2.0 laws, the V2.1 state-dependent leakage, and the V2.2 ledger and safe rule. It adds the four source behaviours and the H-horizon actor, and reports the audit and the regeneration experiment. The naive Model D is shown insufficient; the next revision should implement a threshold-aware burden before any further scaling.